Table 2. Recovery Accuracy on Real-World Images

TP / FP are distinct (deduplicated) counts. Precision = TP / (TP + FP). Recall = TP / N. F1 = 2PR / (P + R).
Partial renders are counted as FP (conservative scoring).

Dataset	Tool	Total PDF (N)	TP (distinct)	FP (distinct)	Precision	Recall	F1
DFRWS-2007	Scalpel	13	0	12	0	0	-
	PhotoRec	13	2	0	1	0.154	0.267
	Foremost	13	1	0	1	0.077	0.143
	SCAR	13	9	0	1	0.692	0.818
	Cohen (2007) PDF Carver	13	8	11	0.421	0.615	0.5
	Chen et al.	13	4	1	0.8	0.308	0.444
streamxref_0100MB	Scalpel	20	11	9	0.55	0.55	0.55
	PhotoRec	20	9	1	0.9	0.45	0.6
	Foremost	20	9	1	0.9	0.45	0.6
	SCAR	20	14	3	0.824	0.7	0.757
	Cohen (2007) PDF Carver	20	0	0	-	0	-
	Chen et al.	20	9	2	0.818	0.45	0.581

A dash (-) marks an undefined value: precision is undefined when no files are recovered (TP + FP = 0), and F1 is undefined when precision or recall is 0.

Table 3. Recovery Accuracy on NIST CFReDS (L0–L5)

Six difficulty levels. TP / FP are distinct (deduplicated) counts. Precision = TP / (TP + FP). Recall = TP / N. F1 = 2PR / (P + R). Partial renders are informational here; add them to FP for conservative scoring.

Dataset	Tool	Total PDF (N)	TP (distinct)	FP (distinct)	Precision	Recall	F1
L0	Scalpel	2	0	2	0	0	-
	PhotoRec	2	2	0	1	1	1
	Foremost	2	2	0	1	1	1
	SCAR	2	2	0	1	1	1
	Cohen (2007) PDF Carver	2	2	0	1	1	1
	Chen et al.	2	2	0	1	1	1
L1	Scalpel	2	0	2	0	0	-
	PhotoRec	2	2	0	1	1	1
	Foremost	2	2	0	1	1	1
	SCAR	2	2	0	1	1	1
	Cohen (2007) PDF Carver	2	2	0	1	1	1
	Chen et al.	2	2	0	1	1	1
L2	Scalpel	2	0	2	0	0	-
	PhotoRec	2	0	0	-	0	-
	Foremost	2	0	0	-	0	-
	SCAR	2	2	0	1	1	1
	Cohen (2007) PDF Carver	2	2	2	0.5	1	0.667
	Chen et al.	2	0	0	-	0	-
L3	Scalpel	2	0	2	0	0	-
	PhotoRec	2	0	0	-	0	-
	Foremost	2	0	0	-	0	-
	SCAR	2	1	0	1	0.5	0.667
	Cohen (2007) PDF Carver	2	2	0	1	1	1
	Chen et al.	2	1	0	1	0.5	0.667
L4	Scalpel	2	0	2	0	0	-
	PhotoRec	2	2	0	1	1	1
	Foremost	2	1	0	1	0.5	0.667
	SCAR	2	2	0	1	1	1
	Cohen (2007) PDF Carver	2	2	0	1	1	1
	Chen et al.	2	2	0	1	1	1
L5	Scalpel	2	0	2	0	0	-
	PhotoRec	2	0	0	-	0	-
	Foremost	2	0	0	-	0	-
	SCAR	2	2	0	1	1	1
	Cohen (2007) PDF Carver	2	2	0	1	1	1
	Chen et al.	2	2	0	1	1	1

A dash (-) marks an undefined value, as in Table 2.